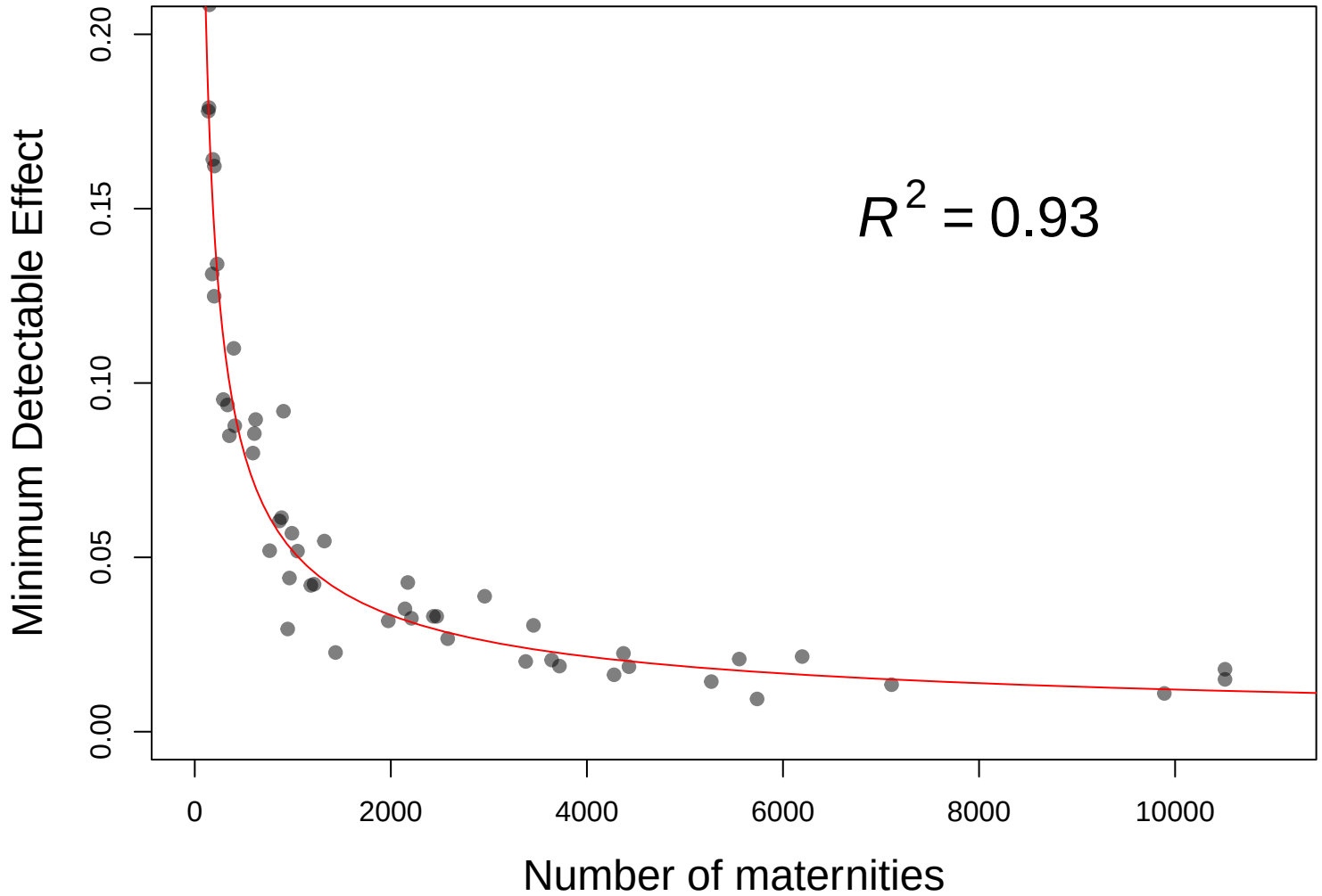
a)

Metric	R^2
records	0.744
maternities	0.93
paternities	0.919
full sibs	0.765
maternal sibs	0.909
maternal half sibs	0.705
paternal sibs	0.905
paternal half sibs	0.722
maternal grandmothers	0.558
maternal grandfathers	0.588
paternal grandmothers	0.583
paternal grandfathers	0.555
maximum pedigree depth	0.137
founders	0.216
mean maternal sibship size	0.554
mean paternal sibship size	0.503
non−zero F	0.256
F > 0.125	0.185
mean pairwise relatedness	−0.008
pairwise relatedness>=0.125	−0.003
pairwise relatedness>=0.25	0.045
pairwise relatedness>=0.5	0.155
var pairwise relatedness	0.045

b)